Sungmin Kim, Ph.D. Candidate

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Research interests:

- Efficient pattern matching algorithms under various equivalence relations
- Subsequences and Simon's congruence
- Decidability problems on formal languages
- Theory of Computation

Education

2022 – Present **Ph.D. candidate in Computer Science, Yonsei University**
Ph.D. Supervisor: Prof. Yo-Sub Han
Expected graduation: August 2026

2017 – 2022 **B.A. in English Language and Literature, Yonsei University**

B.E. in Computer Science, Yonsei University
Double major, Highest honors at graduation.

Research Publications

Journal Articles

1. P. Fleischmann, **S. Kim**, T. Koß, *et al.*, "Matching patterns with variables under Simon's congruence," 2026.
2. Y.-S. Han, **S. Kim**, S.-K. Ko, and K. Salomaa, "Existential and universal width of alternating finite automata," *Information and Computation*, p. 105 337, 2025, invited paper.
3. **S. Kim** and Y.-S. Han, "Approximate Cartesian tree pattern matching," *Theoretical Computer Science*, p. 115 506, 2025.
4. **S. Kim**, S.-K. Ko, and Y.-S. Han, "Simon's congruence pattern matching," *Theoretical Computer Science*, vol. 994, p. 114 478, 2024.
5. **S. Kim**, Y.-S. Han, S.-K. Ko, and K. Salomaa, "On Simon's congruence closure of a string," *Theoretical Computer Science*, vol. 972, p. 114 078, 2023, invited paper.

Conference Proceedings

1. **S. Kim** and Y.-S. Han, "Pattern mining under Simon's congruence," in *Proceedings of the 29th International Conference on Developments in Language Theory (DLT)*, vol. 16036, Springer, 2025, pp. 182–196.
2. **S. Kim** and Y.-S. Han, "Approximate Cartesian tree pattern matching," in *Proceedings of the 28th International Conference on Developments in Language Theory (DLT)*, vol. 14791, Springer, 2024, pp. 189–202.
3. P. Fleischmann, **S. Kim**, T. Koß, *et al.*, "Matching patterns with variables under Simon's congruence," in *Proceedings of the 17th International Conference on Reachability Problems (RP)*, vol. 14235, 2023, pp. 155–170.
4. Y.-S. Han, **S. Kim**, S.-K. Ko, and K. Salomaa, "Existential and universal width of alternating finite automata," in *Proceedings of the 25th International Conference on Descriptional Complexity of Formal Systems (DCFS)*, vol. 13918, 2023, pp. 51–64.

5. **S. Kim**, Y.-S. Han, S.-K. Ko, and K. Salomaa, "On the Simon's congruence neighborhood of languages," in *Proceedings of the 27th International Conference on Developments in Language Theory (DLT)*, vol. 13911, 2023, pp. 168–181.
6. **S. Kim**, Y.-S. Han, S.-K. Ko, and K. Salomaa, "On Simon's congruence closure of a string," in *Proceedings of the 24th International Conference on Descriptional Complexity of Formal Systems (DCFS)*, vol. 13439, 2022, pp. 127–141.
7. **S. Kim**, S.-K. Ko, and Y.-S. Han, "Simon's congruence pattern matching," in *Proceedings of the 33rd International Symposium on Algorithms and Computation (ISAAC)*, vol. 248, 2022, 60:1–60:17.

Under Review

1. **S. Kim** and Y.-S. Han, "Pattern mining under Simon's congruence," invited and submitted to special issue of Information and Computation.
2. **S. Kim** and Y.-S. Han, "Subsequence matching and edit distances under Simon's congruence," submitted to a conference.
3. **S. Kim**, Y.-S. Han, S.-K. Ko, and K. Salomaa, "On the Simon's congruence neighborhood of languages," submitted to Theoretical Computer Science.
4. **S. Kim**, H. Hong, T. Lim, Y.-S. Han, and K. Salomaa, "Decomposing regular languages under shuffle along trajectories," submitted to a conference.
5. **S. Kim**, H. Jin, and Y.-S. Han, "Pattern matching under $\mathcal{R}$-congruence," submitted to a conference.

Academic Activities and Services

2025	Reviewer for STACS 2026
2024 – today	Co-organized Yonsei CS Theory Student Group Seminars
2024	Reviewer for ISAAC 2024
2023 – 2025	Teaching Assistant: Data Structures / Automata and Formal Languages / Theory of Computation / Analysis of Algorithms
2022 – 2025	Problem author, ICPC regional (Asia, Seoul)

Experience

2025 – 2026	Participation in Industry–University Collaborative Project, LG
2024 – 2025	Lab manager for the Theory of Computation Lab (Yonsei University)
Summer 2023	Visiting research at Göttingen University, Germany
Summer 2022	Visiting research at Queen's University, Canada

Miscellaneous

Language Skills

Korean	native speaker
English	fluent

Graduate Scholarships

2022 – 2024	**Accelerated degree program scholarship**

Miscellaneous (continued)

2023 **Academic research support ARF fellowship grant**